

A Deloitte report predicts, “More than 200 million smartphones that can connect with satellite services will be sold in 2024. These phones are expected to contain about US\$2 billion of special chips. In addition, spending on satellite construction and launches could bring the total technology investment to help enable this market to more than US\$3 billion in 2024.”

5G advanced. The first release of 5G Advanced, Release 18, is slated for 2024. It will use technologies like AI and ML, network slicing, analytics and automation, to adapt to the environment, boost performance, manage complex optimisations, and improve sustainability.

It is believed that 5G Advanced will bring mobile broadband to new classes of devices and open 5G to new sectors. It also promises hypersensitive positioning and resilient timing, which can significantly enhance extended reality (XR) and haptic experiences on mobile devices.

Making sure we have a planet to live on

Sustainability. Gartner predicts that by 2027, 25% of CIOs will see their personal compensation linked to their sustainable technology impact. Organisational leaders will start focusing on sustainable tech this year, to ensure positive ESG outcomes in the long term.

Semiconductor sustainability.

With the semiconductor industry expected to rise from the downturn this year, there is going to be more focus on sustainability—not only in greenfield projects but brownfield plants as well. A Deloitte report points out that while the equipment and infrastructure in new plants may be more sustainable due to the use of more efficient materials, and so on, moving from a mature technology such as 28nm manufacturing

to advanced node manufacturing at 2nm needs 3.5 times as much energy, 2.3 times as much water, and emits 2.5 times as much greenhouse gases. Hence, manufacturing transformation in older plants is likely to contribute more to sustainability than the advanced setups in new projects.

Deloitte predicts that there will be a year-over-year decline in average water intensity and energy intensity in 2024, as well as declines in energy intensity and growth in the percentage of energy used by leading chipmakers that is renewable.

Telco sustainability. Most telcos practice sustainability by using as much renewable energy as possible for their operations and adopting electric vehicles (EVs) for their field service fleet. Apart from that, it has been proven that shutting down networks like copper wire and 3G that have too few subscribers to justify the costs, can save a lot of energy.

However, the major savings is expected to come from 5G equipment that can ‘snooze’ or go into a low-power mode when not in use. Some telcos go to the extent of turning off their 5G infra at times of low usage, forcing customers to fall back on 4G networks.

But we are going to see a huge change this year, with Ericsson, Nokia, Samsung, and Huawei introducing 5G remote radio units (RRUs), baseband units (BBUs), and active antenna processing units (AAUs), which are powered by AI algorithms and new semiconductors, and use much less energy than the first generation of 5G gear. The new equipment is expected to lower power consumption by 20-50%, and up to 94% during periods of low traffic.

The EV imperative. Industry leaders like Toyota, Nissan, Honda,

Volkswagen, Ford, and BMW are seriously working on incorporating solid-state batteries in EVs as it could improve driving range and safety, while reducing charging time and manufacturing costs. While some are committed to developing their own solid-state batteries, others are working with external battery manufacturers for this.

The EV industry is also exploring the wireless charging of vehicles—just like the wireless charging pads for your mobile phones. Imagine, you just park your car at the mall, and it gets charged!

A little further into the future, vehicle-to-grid technologies will enable your vehicle to exchange (send and receive) energy with the grid.

Batteries to the rescue. Solid-state batteries are a major paradigm shift for EVs, as they have higher storage capacities, longer life and better safety at a price that could potentially be lower than traditional batteries. “The aim is to support electric mobility and accelerate long-duration energy storage, which is critical to speed-up the energy transition to renewables and the acceleration of smart grids. While LFP (lithium ferro-phosphate) and NMC (nickel manganese cobalt) are becoming standard for electric vehicle applications, several technologies with the chemistry of batteries are being explored, such as cobalt-free (sodium-ion) or solid-state batteries, with a likely acceleration in 2024,” says Capgemini’s report, *Technovision: Top 5 Tech Trends to Watch in 2024*.

Seems to be a good year for the tech world. But, as we said earlier, these are just predictions and could take a tangential turn at any point in time! Differing opinions and unexpected outcomes are part of the game. We can just wait and watch how things turn out. But you can bank on us to let you know! **EFY**